Course Curriculum

Course Code: CSIT805

Course Level PG

Course Title Cloud Infrastructure and Services

Course Description : PG level course

Credit Units

L	T	P/S	SW	AS/DS	FW	No. of PSDA	Total Credit Unit
3.00	0.00	0.00	2.00	0.00	0.00	0	4.00

Course Objectives :

SN.	Objectives
1	To give students the skills and knowledge to understand how Cloud Computing Architecture can enable transformation, business development and agility in an organization.
2	To understand Concepts and infrastructure of cloud computing and its business applications.
3	To understand the role and responsibilities of professional field, how to deal with ethical, legal, security and social issues and responsibilities related to cloud computing.
4	Apply concept of cloud infrastructure and services in developing solutions for problems defined in Viksit Bharat@2047 guidelines document

Pre-Requisites : General

SN.	Course Code	Course Name

Course Contents / Syllabus :

SN.	Module	Descriptors / Topics	Weightage
1	Module I: Journey To the Cloud	Business Drivers, definition, essential characteristics, and phases of journey to the Cloud. Business drivers for Cloud computing, Definition of Cloud Computing, Characteristics of Cloud Computing as per NIST, Steps involved in transitioning from Classic data center to Cloud computing environment.	15.00
2	Module II: Classic Data Center (CDC)	Key elements of CDC- compute, storage, and network, with focus on storage networking, business continuity, and data center management, Application, DBMS, Compute, Storage and Networking, Object based and Unified storage technologies, Business continuity overview and backup, Replication technologies, CDC Management.	15.00
3	Module III: Virtualized Data Center (VDC)	Virtualization of core technologies in a data center, leading to Virtualized Data Center (VDC). It explains the fundamental concepts of compute, storage, networking, desktop and application virtualization. Concepts and techniques employed for ensuring business continuity in a VDC. Compute, Storage, Network virtualization techniques, Virtual Machine components and process of converting physical to VMs, Block and file level storage virtualization, Virtual provisioning and automated storage tiering, VLAN, and VSAN and their benefits, key network traffic management techniques in VDC, Methods for implementing desktop virtualization, their benefits, and considerations, Application virtualization methods, benefits, and considerations, Backup and recovery of Virtual Machines, VM replication and migration technologies, Recovery options from total site failure due to a disaster.	20.00
4	Module IV: Cloud Computing and Infrastructure	Essential characteristics of Cloud Computing, the different Cloud services and deployment models, the economics of Cloud, Cloud infrastructure components, and Cloud service creation processes. Cloud service management processes that ensure that the delivery of Cloud Services is aligned with business objectives and expectation s of Cloud service consumenrs. Cloud services models, Cloud deployment models, Economics of Cloud, Cloud infrastructure components, Cloud service creation processes, Cloud service management processes.	20.00
5	Cloud Security and Migration to Cloud	Security concerns and migration considerations to cloud. Key security concerns and threats and details Cloud model suitable for different categories of users. Security concerns and counter measures in a VDC and Cloud environment, Governance, Risk, and Compliance aspects in Cloud, Cloud security best practices, Cloud models suitable for different categories of users, Considerations for choosing applications suitable for Cloud, Different p	15.00
6	Recent Trends and Developmen	Discussion on all the recent trends and developments in the application of Cloud Computing Services and Architecture. Application of all the concept in development of applications using Cloud Computing for developing projects as per Viksit Bharat@2047 guidelines	15.00

	t		

Course Learning Outcomes :

SN.	Course Learning Outcomes

Pedagogy for Course Delivery :

SN.	Pedagogy Methods
1	The course would be covered under theory and laboratory in blended / flip mode. In addition to assigning project–based learning, early exposure to hands-on design to enhance the motivation among the students. It incorporates designing of problems, analysis of solutions submitted by the student and how learning objectives were achieved. Continuous evaluation of the students would be covered under Self work, project and viva

Theory /VAC / Architecture Assessment (L,T & Self Work): 100.00 Max : 100

Attendance+CE+EE : 5.00+45.00+50.00

SN.	Type	Component Name	Marks
1	Attendance		5.00
2	Attendance		5.00
3	Attendance		5.00
4	End Term Examination (OMR)		50.00
5	End Term Examination (OMR)		50.00
6	End Term Examination (OMR)		50.00
7	Internal	CLASS TEST	20.00
8	Internal	VIVA VOCE	5.00
9	Internal	Assignment	10.00
10	Internal	PROJECT	10.00

Lab/ Practical/ Studio/Arch. Studio/ Field Work Assessment : 0.00 Max : 100

N/A

List of Professional skill development activities :

No.of PSDA : 5

SN.	PSDA Point
1	Survey and find benefits of Cloud Services, also state the different application areas
2	Survey and find top companies offering job opportunities, and technical question bank
3	Take any real-life problem and prepare a solution of the same
4	Present above mentioned activities report in the class
5	Develop project using all concept learnt for solving any one problem for Viksit Bharat@2047 vision

Text & References :

SN.	Type	Title/Name	Description	ISBN/ URL
1	Book	Cloud Computing: A Practical Approach Author: Anthony T. Velte, Publisher: Tata Mcgraw Hill Educa		0070683514
2	Book	Cloud Computing for Dummies Author: Halper Fern, Kaufman Marcia, Bloor Robin, Hurwit Judith, Publi		8126524871
3	Book	• Gautam Shroff: Enterprise Cloud Computing Technology Architecture Applications		978-0521137355
4	Book	Toby velte, Anthony Velte, Robert Elsenpeter; Cloud computing, A practical Approach		0071626948
5	Book	Dimitris N. Chorafas, Cloud Computing Strategies		1439834539
6	Url	http://www.siteground.com/tutorials/cloud/cloud_computing_infrastructure.htm		
7	Url	http://thecloudtutorial.com/		
8	Book	Cloud Computing: A Practical Approach Author: Anthony T. Velte, Publisher: Tata Mcgraw Hill Educa		0070683514
9	Book	Cloud Computing for Dummies Author: Halper Fern, Kaufman Marcia, Bloor Robin, Hurwit Judith, Publi		8126524871
10	Book	• Gautam Shroff: Enterprise Cloud Computing Technology Architecture Applications		978-0521137355
11	Book	Toby velte, Anthony Velte, Robert Elsenpeter; Cloud computing, A practical Approach		0071626948
12	Book	Dimitris N. Chorafas, Cloud Computing Strategies		1439834539
13	Url	http://www.siteground.com/tutorials/cloud/cloud_computing_infrastructure.htm		
14	Url	http://thecloudtutorial.com/		
15	Book	Cloud Computing: A Practical Approach Author: Anthony T. Velte, Publisher: Tata Mcgraw Hill Educa		0070683514
16	Book	Cloud Computing for Dummies Author: Halper Fern, Kaufman Marcia, Bloor Robin, Hurwit Judith, Publi		8126524871
17	Book	• Gautam Shroff: Enterprise Cloud Computing Technology Architecture Applications		978-0521137355
18	Book	Toby velte, Anthony Velte, Robert Elsenpeter; Cloud computing, A practical Approach		0071626948
19	Book	Dimitris N. Chorafas, Cloud Computing Strategies		1439834539
20	Url	http://www.siteground.com/tutorials/cloud/cloud_computing_infrastructure.htm		
21	Url	http://thecloudtutorial.com/		
22	Book	Cloud Computing: A Practical Approach Author: Anthony T. Velte, Publisher: Tata Mcgraw Hill Educa		0070683514
23	Book	Cloud Computing for Dummies Author: Halper Fern, Kaufman Marcia, Bloor Robin, Hurwit Judith, Publi		8126524871
24	Book	• Gautam Shroff: Enterprise Cloud Computing Technology Architecture Applications		978-0521137355
25	Book	Toby velte, Anthony Velte, Robert Elsenpeter; Cloud computing, A practical		0071626948

		Approach		
26	Book	Dimitris N. Chorafas, Cloud Computing Strategies		1439834539
27	Url	http://www.siteground.com/tutorials/cloud/cloud_computing_infrastructure.htm		
28	Url	http://thecloudtutorial.com/		
29	Book	Cloud Computing: A Practical Approach Author: Anthony T. Velte, Publisher: Tata Mcgraw Hill Educa		0070683514
30	Book	Cloud Computing for Dummies Author: Halper Fern, Kaufman Marcia, Bloor Robin, Hurwit Judith, Publi		8126524871
31	Book	• Gautam Shroff: Enterprise Cloud Computing Technology Architecture Applications		978-0521137355
32	Book	Toby velte, Anthony Velte, Robert Elsenpeter; Cloud computing, A practical Approach		0071626948
33	Book	Dimitris N. Chorafas, Cloud Computing Strategies		1439834539
34	Url	http://www.siteground.com/tutorials/cloud/cloud_computing_infrastructure.htm		
35	Url	http://thecloudtutorial.com/		
36	Book	Cloud Computing: A Practical Approach Author: Anthony T. Velte, Publisher: Tata Mcgraw Hill Educa		0070683514
37	Book	Cloud Computing for Dummies Author: Halper Fern, Kaufman Marcia, Bloor Robin, Hurwit Judith, Publi		8126524871
38	Book	• Gautam Shroff: Enterprise Cloud Computing Technology Architecture Applications		978-0521137355
39	Book	Toby velte, Anthony Velte, Robert Elsenpeter; Cloud computing, A practical Approach		0071626948
40	Book	Dimitris N. Chorafas, Cloud Computing Strategies		1439834539
41	Url	http://www.siteground.com/tutorials/cloud/cloud_computing_infrastructure.htm		
42	Url	http://thecloudtutorial.com/		